

Assessment Task 3: Presentation & Technical Interview (40%).

This assignment was locked Sep 15 at 23:59.

Course code and name: COSC2753 Machine Learning

Length: Oral presentation within 10 minutes + 02 minutes for demonstration.

Submission: PPT/PPTX file + one video recording your presentation before the oral presentation day.

Type: Group & Individual

Feedback mode: Written feedback

Late work: + Not allow. The slides and video must be submitted one day before the first presentation day of the class.

+ No submission will lead to no presentation.

Learning Objectives Assessed

- **CLO1:** Understand the fundamental concepts and algorithms of machine learning and applications
- **CLO2:** Understand a range of machine learning methods and the kinds of problem to which they are suited
- **CLO5:** Understand major application areas of machine learning
- **CLO6:** Understand the ethical considerations involved in the application of machine learning

Ready for Life and Work

Completing this assessment would help you achieve the following employability skills:

- Analyse and model requirements and constraints for the purpose of designing and implementing solutions to a learning challenge
 - Evaluate and compare approaches and algorithms on the basis of the nature of the problem/task being addressed
 - Interpret abstract theoretical propositions, choose methodologies, justify conclusions and defend professional decisions to both IT and non-IT personnel via technical reports of a professional standard and technical presentations
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Assessment Details

Task

Using machine learning in real-world settings involves more than just running a data set through a particular algorithm. You need to have skill in researching and understanding what others have done to solve similar problems. In this assignment, you will do a critical analysis of a ML project, as well as improvements/extensions that could be made based on a literature review of the state-of-the-art approaches. You will also demonstrate theoretical understanding of ML techniques.

The projects you are to investigate for this assignment is as the project you do in Assignment 2.

A. Presentation: you are required record a presentation less than 10 minutes long that includes the following:

- (approximately 2 minutes) The introduction and a review of the problems that you want to solve. It should include the techniques in literature that are used to solve same/similar problems.
- (approximately 3 minutes) Present your approach, compare and contrast yours to the approaches found in your review.
- (approximately 3 minutes) Present the results of your experiments, a critical analysis of the model(s) that you have developed, and the comparison must be included.
- (approximately 2 minutes) Discuss how the project can be extended to cover some challenging scenarios/limitations.

You may use slides in your presentation. However, content on the slides that are not adequately explained (verbally) would not receive any marks. The instructions for recording a presentation is provided. Please follow the instructions carefully and make sure the access privileges are configured correctly so that the markers can access the recording.

Incorrectly submitted/corrupted recordings or recordings that cannot be accessed will receive zero marks. If the recorded video is over 10 minutes long. Only the part up to 10 minutes will be accessed. The over length part of the recording will be ignored.

All students are required to appear in the recordings, including showing their faces for identification purposes. Recordings without students' faces will not be evaluated and will be considered as incomplete.

B. Demonstration: After your group presentation, you are allowed to spend from 02 - 04 minutes for your application/model demonstration. You should show your main functionalities of your applications regarding to all tasks that you are required to solve. If the time is left, you can show other functions that are relevant too. The length of demonstration is not counted for the presentation, but not over 04 minutes.

Instruction for important content that you should include in your presentation is as following:

1. Review

You should conduct a review to identify the following items and discuss your findings in the presentation. Your review should identify:

Other works in literature that solves the same/similar problem as the one you have solved.

Other techniques and algorithms that can be used to solve the problem given to you.

A good literature review:

- Should follow a logical structure.
- Should not just provide a list of related papers. Papers should be discussed in relation to why are they relevant for this problem, what is good about them and what are the limitations.
- Should discuss literature from peer-reviewed sources. Wikipedia or web discussion forums are not considered as peer-reviewed sources.

Remember that good literature review provides factual statements that summaries the work in literature in a way that is useful for the listener to understand the context and follow your rationale. Statements such as: "Ruwan did <xyz>. Andrew did <xyz>. Wei did <xyz>" is not a literature review. This is an annotated bibliography. Instead, you should aim for statements such as: "To solve the problem <xyz> Ruwan did <xyz>, while this method can handle <xyz> it has the limitations <xyz>. To overcome this limitation, Andrew proposed <xyz>..."

2. Your approaches

Your presentation should include the approach you use for each tasks, and a critical analysis of the solution that you developed in your Assignment 2. This part may include:

- A brief summary of your solution to the problem.
- Other approaches considered while developing the solution.
- Compare your approaches with other approaches found in your review.
- Why do you think the approach is adequate to solve the problem? Any requirements for your approaches?

3. Results & Critical Analysis

- How you set up the environments for experiments.
- What is the results? Analyse the models' outputs.
- The results is aligned with the your assumption when choosing approaches or not? Anything is new and un-expected?
- Limitations of the model you developed? How does it affect to the results/performance?

4. Extensions

In this section, you should explore how your system could be extended to address a specific scenario or application in future. Your discussion should include a clearly defined problem formulation and a well-reasoned approach to tackling it. Additionally, explain how you would evaluate whether the proposed solution effectively addresses the new task. Suggestions for future improvements or directions for further development are strongly encouraged.

Support Resources

This assessment requires that you meet RMIT's expectations for academic integrity. More information and advice on how to avoid plagiarism are available in the Getting Started module.

Open [the academic integrity page](#).

Additional library and learning resources are available to help with the assessment in this course

Link to [Assignment Support](#).

Submission Instruction

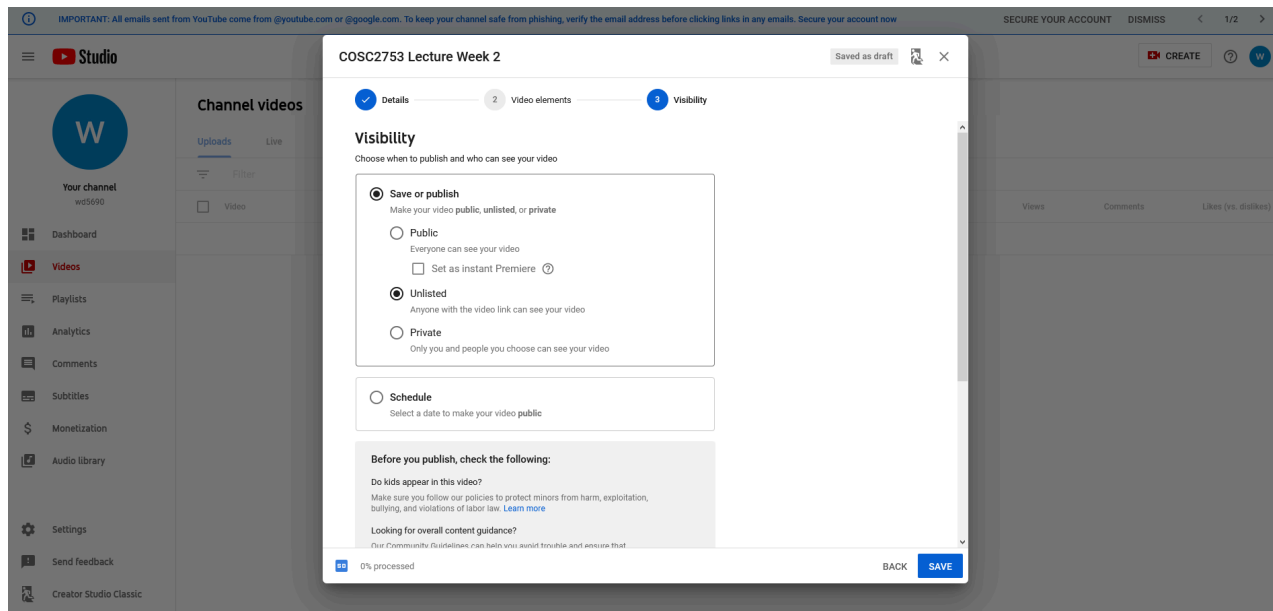
In this assignment, you will submit two things:

- **The first submission:** the presentation (ppt/pptx/canva link) file.
- **The second submission:** the video or the link to the online video recording of your presentation, which can be uploaded and hosted on any cloud-based storage, as long as you set up the right view access so that the teaching team can watch it with a good quality.
- Only **one student from your group will submit both two things.**

There are different cloud storage that you may use, including Google Drive, YouTube, or OneDrive. The following instructions is for recording and sharing the online video by using Teams.

1. Make sure to set the privacy settings of your uploaded video to allow people with the link can watch/view it (*note*: when uploading to YouTube, you need to select the option "Unlisted" for your video, in the "Visibility" section - see the figure below)
2. Copy the link and put it on a document (doc, docx, pdf, txt)
3. Submit the document with the link embedded inside it to Canvas

Note: make sure to choose the "Unlisted" option if you choose to upload your video to YouTube



Rubric

You can review the rubric for this assignment before making a submission to Turnitin.

Criteria	Ratings					Pts
Review	20 to >16.0 pts HD Outstanding across the course. Content presented in a logical order and the explanations of literature is correct. The discussion on papers given is excellent and extremely thorough. Covers both required aspects and the discussions makes careful consideration of the unique aspects of the given ML problem. In addition to the given papers, at least TWO more papers that are closely related to the project are identified. Discussion on additional papers is excellent and extremely thorough. The literature relevant to each task must be thoroughly reviewed and discussed.	16 to >14.0 pts DI Content presented in a logical order and the explanations of literature is correct. The discussion on papers given is good and reasonably thorough. Covers both required aspects and the discussions makes careful consideration of the unique aspects of the given ML problem. In addition to the given papers, at least ONE additional papers that are closely related to the project are identified and explained well. Discussion on additional papers is good and reasonably thorough. The literature relevant to each task must be thoroughly reviewed and discussed.	14 to >12.0 pts CR Content presented in a logical order and the explanations of literature is sometimes incomplete or may contain small discrepancies. The discussion on papers given is sufficient. Covers both required aspects and the discussions makes careful consideration of the unique aspects of the given ML problem. No additional papers are identified or, the papers identified are not related or, the explanations on the additional papers are not adequate. The literature relevant to each task must be thoroughly reviewed and discussed.	12 to >10.0 pts PA Content presented in a logical order and the explanations of literature incomplete or may contain many discrepancies. The discussion on papers given is minimally sufficient. Covers both required aspects and the discussions makes minimal consideration of the unique aspects of the given ML problem. No additional papers are identified or, the papers identified are not related or, the explanations on the additional papers are not adequate. The literature relevant to each task must be thoroughly reviewed and discussed.	10 to >0 pts NN Content not presented in a logical order and the explanations of literature is incomplete. Poor, superficial, or incomplete review that does not meet the minimum requirements for PA. No additional papers are identified or, the papers identified are not related or the explanations on the additional papers are not adequate. The literature relevant to one or some tasks has not been thoroughly reviewed and discussed.	20 pts
Critical Analysis	20 to >16.0 pts HD Outstanding across the course. The critical analysis on proposal method is excellent and clearly discuss all the elements mentioned in the specifications. Limitations of the model identified. Evidence based discussion. All models have been thoroughly investigated.	16 to >14.0 pts DI The critical analysis on proposal method is good and clearly discuss most of the elements mentioned in the specifications. Limitations of the model identified. Evidence based discussion. All models have been thoroughly investigated.	14 to >12.0 pts CR The critical analysis on proposal method is good and clearly discuss most of the elements mentioned in the specifications. Justifications given to some elements are limited or may not present evidence. Limitations of the model identified. Minimal evidence provided. All models have been thoroughly investigated.	12 to >10.0 pts PA The critical analysis on proposal method is minimally sufficient and discuss some of the elements mentioned in the specifications. Justifications not given or inaccurate. Limitations of the model identified. All models have been thoroughly investigated.	10 to >0 pts NN The critical analysis on proposal method is not adequate and missing discussions of the elements mentioned in the specifications. Justifications not given or inaccurate. Limitations of the model not identified or inadequate. Not all models have been thoroughly investigated.	20 pts

Criteria	Ratings					Pts
Extension	20 to >16.0 pts HD Outstanding across the course. Presents a clearly defined and well-justified extension of the system. The proposed approach is innovative, highly relevant, and demonstrates deep understanding. Evaluation methods are thorough and appropriate. Future directions are insightful and well-articulated. The proposed solution is excellent and has considered all the required aspects. The justifications of the elements of the proposed solution is excellent and explained well. Clearly identified how to measure success.	16 to >14.0 pts DI Clearly defines a relevant extension with solid justification. The approach is thoughtful and mostly well developed. Evaluation strategy is appropriate and future work is reasonably discussed, with some insight. The proposed solution is good and has considered most of the main required aspects. The justifications of the elements of the proposed solution is good and explained well. Clearly identified how to measure success.	14 to >12.0 pts CR Provides a reasonably clear extension idea with some justification. The proposed method is adequate but lacks depth or originality. Evaluation methods are mentioned but not well developed. Future directions are basic or limited. The proposed solution is sufficient and has considered most of the main required aspects. There are minor errors in the formulation. The justifications of the elements of the proposed solution is sufficient and explained well. Clearly identified how to measure success.	12 to >10.0 pts PA Extension idea is vaguely presented or lacks strong relevance. Approach is simplistic or incomplete. Evaluation is superficial or unclear. Future work is minimal or generic. The proposed solution is minimally sufficient and has considered some of the main required aspects. There are errors in the formulation. The justifications of the elements of the proposed solution is minimally sufficient and explained well. Does not identify how to measure success.	10 to >0 pts NN No meaningful extension proposed, or extension is unclear, irrelevant, or unjustified. Approach and evaluation are missing or poorly conceived. No future directions provided. Incomplete or error ridden solution to the extension. Does not identify how to measure success.	20 pts
Demonstration	20 to >16.0 pts HD Demonstration is exceptionally clear, complete, and technically sound. The solution is fully functional and showcases advanced understanding and implementation of ML models. Clear linkage between model, data preprocessing, metrics, and results. Excellent visualizations and interpretability methods are used. All aspects are explained with clarity. Demo all tasks correctly.	16 to >14.0 pts DI Demonstration is clear and complete without issues. The implementation is correct and reflects strong understanding. Results are well-presented with appropriate metrics and visualizations. Some explanation may lack depth but is mostly well-articulated. Demo all tasks correctly.	14 to >12.0 pts CR Demonstration is mostly functional but may lack clarity or depth in explanation. Key aspects are implemented correctly, though minor mistakes or omissions may be present. Visualization and analysis are somewhat helpful but not thorough. Demo all tasks correctly.	12 to >10.0 pts PA Demonstration is basic and may include errors or incomplete elements. Understanding of the model and results is superficial. Limited use of evaluation metrics or visual aids. Explanation is minimal or vague. Demo all tasks correctly.	10 to >0 pts NN Demonstration is largely incomplete, incorrect, or missing. Little or no evidence of understanding of the machine learning workflow. Poor or no explanation. One or some tasks can not be demo correctly.	20 pts
Presentation & QA	20 to >16.0 pts HD Presentation is highly professional, well-structured, and	16 to >14.0 pts DI Presentation is clear, logically structured, and mostly engaging.	14 to >12.0 pts CR Presentation is generally understandable with	12 to >10.0 pts PA Presentation is hard to follow or poorly organized. Visual aids	10 to >0 pts NN Presentation is unclear or disorganized. Visuals	20 pts

Criteria	Ratings					Pts
	engaging. Content is delivered clearly with excellent use of visuals. All team members contribute meaningfully. During Q&A, responses show deep understanding, critical insight, and confidence. Manage time precisely.	Visuals support the narrative well. Q&A responses demonstrate good understanding. There are one or some points not clear or may lack deeper insight in some areas. Manage time precisely.	some organizational or delivery issues. Visuals are basic. Q&A responses are mostly correct but may lack clarity or depth. Manage time precisely.	are minimal or ineffective. Q&A responses are limited or indicate only a basic understanding of the topic. Not manage time precisely.	are missing or ineffective. Little to no participation in Q&A or responses indicate lack of preparation or understanding.	
Total Points: 100						